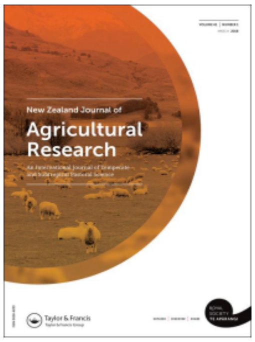

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Depth distribution of simulated urine in a range of soils soon after deposition

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Abstract The distribution of a solution simulating cow urine in soil was measured within 6 hours of deposition, by applying 2 litres of KBr-pyranine dye solution to the soil surface at a rate of 0.2 litres s^{-1}, followed by excavation and sub-sampling of the area affected by urine in 10-cm depth increments until no further pyranine dye could be detected under UV fluorescence. Preferential flow of simulated cow urine to below 20 cm soil depth occurred in 9 of the 10 soils examined. Up to 68% of applied solution moved below 20 cm, at an average of 17% over all soils examined. Two soils exhibited a high degree of preferential flow and excluding these from the data reduced the average loss to 11%. Although this preferential flow was significant, much of the urine nutrients moving via preferential flow would still be available for uptake by plant roots at depth, and therefore cannot be considered a loss to pasture. The movement of urine to below 20 or 30 cm soil depth was best predicted by saturated hydraulic conductivity calculated as the mean for the 0–30 cm soil layer. Saturated hydraulic conductivity measurements made for any one 5-cm depth increment, particularly the 0–5 cm soil depth, were poor predictors of urine flow to below 20 or 30 cm depth.

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Keywords preferential flow; cow urine; saturated hydraulic conductivity;potassium; pasture root distribution

INTRODUCTION

Much of the dietary potassium (K) ingested by grazing cows is excreted, with only 5–10% being retained in milk and meat product (Hutton et al. 1967). Most of the K excreted is in the form of urine, with only 10–30% being excreted in the form of faeces (Hutton et al. 1967; Betteridge et al. 1986; Barrow 1987; Haynes & Williams 1993). This large and random return of immediately available urine K to pastures represents a potential inefficiency in the recycling of K between soil, plants, and animals. The amount of urinary K deposited within the urine patch is often far in excess of immediate plant requirements and has been estimated as being equivalent to an application of approximately 1000 kg K ha^{-1} (Saunders 1984; Williams et al. 1989). Little is known of the exact fate of K in urine patches: luxury plant uptake, exchange reactions with soil colloids, and leaching are all processes which may act as sinks for soil solution K when it is present in such high concentrations.

Losses of K by leaching are thought to occur primarily as a result of macropore flow of urine to below the plant root layer (Williams et al. 1990a). As the volume of urine voided by cattle is much greater than that for sheep, macropore flow is greater following cattle urinations than those of sheep (Williams & Haynes 1994). Large and rapid losses of urine K have been measured from topsoil cores treated with dairy cow urine (Williams et al. 1990b, 1990c), with <1–74% of urine K moving preferentially through soil macropores to below 15 cm soil depth. In a field study examining seven soils from Manawatu, Taranaki, and Waikato, New Zealand, Williams et al. (1990a) also observed macropore flow losses of 0–48% of the urine K applied, estimated from the non-recovery of urine Br^- in the 0–15 cm soil layer. In the studies using intact topsoil cores, losses of K in subsequent drainage, induced by applying simulated rainfall, were much less important than macropore flow losses. This preferential movement of urine K to below 15 cm was found to be more dependent on soil surface physical properties, indicated by the significant relationship between urine patch surface area and Br retention in the topsoil layer, than soil chemical and mineral characteristics (Williams et al. 1990a).

Although macropore flow to below plant rooting depth has been identified as a significant loss mechanism, field leaching studies have, in contrast, found relatively small losses of K in drainage waters measured at greater than 45 cm depth. Annual drainage losses of 6–10 kg K ha^{-1} have been noted for intensively grazed dairy pastures not receiving N fertiliser in both Waikato and Southland, New Zealand (Rajendram et al. 1998). In the Waikato study, K losses remained relatively constant as N fertiliser inputs increased up to a rate of 400 kg N $ha^{-1}yr^{-1}$. Under conditions of higher rainfall and drainage, Steele et al. (1984) measured slightly higher drainage losses of 14 kg K ha^{-1} from Northland pastures grazed by steers. Leaching losses of K from simulated urine (55 g K m^{-2}) applied to large lysimeters have been measured at 1.8% of that applied (Early et al. 1998). Low drainage losses of K have also been reported for sheep-grazed pastures (Heng et al. 1991; Sakadevan et al. 1993) and for areas of pasture treated with either sheep or cattle urine (Williams & Haynes 1994).

The apparent discrepancy between the amount of K lost in drainage waters and that lost via preferential flow to below plant rooting depth may be attributable to the immobilisation of urinary K at depth, or perhaps to plant uptake from below 15 cm soil depth. There may also be the possibility that macropore flow of urine through small (15–20 cm deep) cores is over-estimated due to edge effects or the flow of urine through cavities created because of soil disturbance during extraction of the core from the soil, or because a closed macropore ending just below the sample depth becomes a free-draining pore. Williams et al. (1990b, 1990c) did note, however, that no edge-flow occurred in the small lysimeters used in their studies. To address these questions for a wider range of soil types, a series of field experiments was conducted in Waikato, Canterbury, and Southland, New Zealand, to directly measure the distribution of simulated urine in soil soon (within 6 hours) after its deposition. Sites examined included on-going leaching experiments in Waikato and Southland where drainage losses of K are being measured as well as the depth-distribution of exchangeable K.

The work reported here seeks to relate macropore flow of urine K to various soil depths, as measured in the field, to specific measurements of soil physical conditions such as saturated hydraulic conductivity and air permeability. Our hypothesis is that the movement of K in cattle urine to below plant rooting depth by macropore flow is dependent on the physical properties of the soil. Identification of a simple, consistent relationship between macropore flow losses of K and soil physical properties would significantly improve the accuracy of current K fertiliser recommendation models.

MATERIALS AND METHODS

Sites

The distribution of urine through the soil profile was examined in 10 soils commonly used for dairy production. A list of the soils and their classification is given in Table 1. At the Fleming, Pukemutu, Silverdale, Whangaripo, Te Kowhai, and Taupo sites, the experiment was conducted twice, nominally when soils were "wet" (close to Field Capacity) and then again when "dry" (a mid-summer sampling). For two Southland soils, Fleming and Pukemutu, there was little difference in soil moisture levels between the "wet" and "dry" samplings because of a wet summer; data for wet and dry samplings were therefore combined. Two measurement sites were used for the Fleming soil, one an undrained site and the other a mole- and tile-drained site immediately adjacent to the undrained site.

Urine application and soil sampling

Three replicate areas of pasture were chosen at random at each site. Herbage was clipped to approx. 5 cm before each experiment. Simulated urine patches were created by the application of 2 litres of solution containing KBr (20 g l^{-1}) to which pyranine dye (4 g l^{-1}) was added as a fluorescent tracer. Each urination event was simulated by pouring the solution onto pasture from a height of 1 m and at a rate of 0.2 l s^{-1} (Williams et al. 1990a). Although it is recognised that, due to its limited reaction with soil mineral surfaces, Br has a greater potential for movement to depth than K, for the purposes of this study Br was assumed to be a good indicator of the movement of urine K in soil soon after deposition. The results presented here therefore represent maximum potential values for K movement in urine-affected soil.

For the five Southland sites (four soils) examined, the surface area of each urine patch was marked onto tracing paper 20 minutes after urine deposition. This interval was sufficient to allow for any ponded simulated urine to infiltrate the soil before tracing commenced. The surface area of the urine patch was identified by the fluorescing of the pyranine dye under a UV lamp. A black polythene sheet was erected as a tent over the urine patch to shield out any sunlight and improve shape definition. Following the marking of the edge of the urine patch, a large rectangle was marked out to completely encompass the area of soil affected by the urine, with the edge of the rectangle no closer than 5–10 cm from the outer edge of the urine patch. Within 6 hours of urine application, soil was excavated from the rectangle in 10-cm depth increments, thoroughly mixed for each depth, weighed, and sub-sampled in 5-fold replication for later Br analysis. Three replicate bags of soil were also taken from each soil layer for soil moisture content determination.

The pyranine dye was used to identify how far the urine had moved, as well as to identify any obvious macropore flow via earthworm channels or fissure planes. Thus, following the excavation of each soil layer, the bottom and sides of the rectangular hole were examined for the presence or absence of dye. In some cases rectangles were excavated to a total depth of 80 cm before total absence of dye was recorded. The lateral spread of urine was minimal, and very few holes needed widening for recovery of all dye and Br. For the base of each soil layer excavated, features of soil structure and the pattern of dye movement were recorded, including abundance of any noticeable large macropores, cracks, mottling, etc., and the distribution of pyranine dye in relation to these.

Preliminary samplings indicated that an extremely thorough mixing of each soil layer was required to quantitatively recover the applied Br, particularly for the 0–10 cm soil layer. The reason for this was because much of the applied Br was retained in the plant root mat in the surface 0–5 cm soil layer. To thoroughly mix each soil layer, peds were crumbled by hand until soil aggregates were less than approximately 3 mm diam. Crumbled soil was then thoroughly mixed by spade and shovel on a large canvas sheet before sub-samples were taken for Br and soil moisture determination.

Table 1 Some physical properties of the 10 soils examined for the distribution of urine Br. New Zealand Soil Classification from Hewitt (1992).

Site no.	Soil name	Location	NZ Soil Classification	Texture	Drainage status
1	Waikiwi	Southland, South Island	Typic Firm Brown Soil	silt loam	well drained
2	Pukemutu	Southland, South Island	Perch-Gley Pallic Soil	clay loam	poorly drained
3	Mokotua	Southland, South Island	Mottled Firm Brown Soil	silt loam	well drained
4	Fleming	Southland, South Island	Mottled Fragic Pallic Soil	silt loam	artificially drained
5	Fleming	Southland, South Island	Mottled Fragic Pallic Soil	silt loam	poorly drained
6	Temuka	Canterbury, South Island	Typic Perched-gley Melanic Soil	silt loam	poorly drained
7	Lismore	Canterbury, South Island	Pallic Orthic Brown Soil	silt loam	well drained
8	Silverdale	Waikato, North Island	Mottled Orthic Brown Soil	silt loam	moderately well drained
9	Whangaripo	Waikato, North Island	Mottled Yellow Ultic Soil	clay	imperfectly drained
10	Te Kowhai	Waikato, North Island	Typic Orthic Gley Soil	silt loam	poorly drained
11	Taupo	Waikato, North Island	Immature Orthic Pumice Soil	sandy loam	well drained

Soil physical properties

Soil hydraulic conductivity (K_s) was determined for each of the soil layers examined. This involved inserting 3 replicate metal cores, 10 cm diam. × 6.5 cm deep, into the soil immediately adjacent to the excavated urine patch. Each core was very gently carved into the soil until the exposed soil surface was 1.5 cm from the top of the core. A very thin (1–2 mm) layer of petroleum jelly was smeared around the inside wall of the metal core, to both ease insertion and fill any cavities between soil and the wall of the core which may have been created during core insertion, thus preventing edge-flow effects. Soil cores were then transported to the laboratory before having their bases "peeled" with gypsum so as to re-open any smeared soil pores. The top surface of each core was also carefully picked with a sharp knife, again to re-open any pores which had been blocked whilst obtaining the cores.

Soil air permeability (K_a) measurements were made for the six South Island soils studied, the Fleming (drained and undrained), Pukemutu, Waikiwi, Mokotua, Temuka, and Lismore soils, immediately prior to the hydraulic conductivity measurement on the same cores. This test was carried out at field moisture content. The sampling ring containing the soil core was placed on steel mesh to allow a free flow of air from the base of the core. A steel plate attached to a hydraulic ram then sealed the surface of the sampling ring. Air was passed from a compressed air cylinder through a tube in the plate to the surface of the soil. Air pressure was measured within the head-space above the soil, but within the sampling ring, by a pressure transducer. Air flow rate and pressure values were used to calculate air permeability using the air flux equation (Corey 1986).

The samples were then placed in a hydraulic conductivity apparatus consisting of a mariotte system connected directly to a computer. Water flow was measured using pressure transducers and the computer software calculated hydraulic conductivity using the Darcy equation (Hillel 1980). Measurements of hydraulic conductivity were made over a 15–20 minute period, with the soils beginning each measurement period at their field moisture content.

Soil bulk density was determined for each soil layer examined, from one core 7.5 cm diam. × 10 cm deep taken from the soil immediately adjacent to the excavated urine patch.

Bromide analysis

Field-moist soil (40 g) was shaken in a 1:5 extract of 0.1 *M* $NaNO_3$ for 30 minutes. Extracts were then allowed to settle for 24 hours before bromide concentrations were determined using a bromide ion-specific electrode (Orion 9435). To overcome some of the variability recorded from the Mokotua and Waikiwi soils, the amount of soil extracted was increased to 250 g. Soil Br concentrations were then calculated on an oven-dry weight basis following gravimetric water content determination.

RESULTS AND DISCUSSION

Br recoveries and depth distributions

From some of the initial urine patches examined, it became evident that total soil Br recoveries were quite variable. As an extreme example, total Br recoveries in the first set of 3 urine patches examined in the Pukemutu soil ranged from 41 to 152%. The cause of this variability was later found to be due to non-representative sub-sampling of each of the soil layers examined, particularly the 0–10 cm layer. Much of the urine solution appeared to be retained within the root mat of this layer, and obtaining a representative sub-sample required the complete destruction of this turf followed by an extremely thorough mixing of the turf with the remaining soil from the layer. When this procedure was adopted, Br recoveries became much more consistent and closer to 100% of that applied. There was seldom complete recovery of Br, with most soils having mean Br recoveries of 87–111% (Table 2). This deviation from 100% of Br applied was assumed to be due to sampling error introduced during the field-weighing and laboratory-analysis of soil. Lateral and vertical movement of urine solution out of the excavated site was discounted: the pyranine dye proved to be a very good visual tracer of urine movement and such movement would have been clearly discernible during pit excavation and inspection.

Thus, to evaluate the depth-distribution of applied urine between sites, Br recoveries for each soil depth were normalised so that the total recovery from each soil pit was 100%. The proportion of applied Br passing below 20 cm and, thus, below the plant rooting zone, was then calculated (Table 3). The proportion of Br moving to below 20 cm ranged from 0 to 68% of that applied. Large losses below 20 and 30 cm were noted for the Whangaripo and Te Kowhai soils, particularly under dry soil conditions. If these two soils are excluded, Br losses below 20 cm ranged from 0 to 25% with an average loss of 11%. The maximum depth to which Br moved was 80 cm, recorded for the wet and dry samplings of the Te Kowhai soil. Br was also recovered from 60–70 cm in the Mokotua, Whangaripo, and Silverdale (dry) soils. Transport of Br to such depths clearly demonstrates macropore flow of urine solution. Indeed, with the exception of the Taupo soil, where piston flow was more evident, Br recoveries in all of the soils examined indicated some preferential flow of applied solute through soil macropores.

Interestingly, although Br was transported to a greater depth in the drained than the undrained Fleming soil, the amount of Br passing below both 20 and 30 cm was less in the drained soil. Observations of pyranine dye in the drained soil clearly revealed the preferential flow of some urine through soil fissures towards the mole drainage channels at this site. The Br data in Table 3, however, suggest that the amount of Br transported via these fissures was actually quite small, based on the greater Br recoveries at 0–20 cm in the drained soil.

The preferential flow observed here is broadly similar to that found for various North Island soils (Williams et al. 1990a, 1990c) and for a stony Lismore silt loam soil (Williams et al. 1990d). In the latter studies, urine solution moving below 15 cm was considered to be unavailable for plant uptake and therefore "lost". In this study we have used 20 and 30 cm as reference depths for the

Table 2 Mean bromide recoveries for each soil.

Soil	Mean %Br recovered	Range (n)
Waikiwi	96	91–102 (3)
Pukemutu	95	41–152 (6)
Mokotua	93	91–95 (2)
Fleming – drained	93	76–133 (6)
Fleming – undrained	89	68–115 (6)
Temuka	101	62–140 (2)
Lismore	98	97–99 (2)
Silverdale – wet	72	49–94 (2)
Silverdale – dry	111	109–113 (2)
Whangaripo – wet	106	102–111 (2)
Whangaripo – dry	103	98–108 (2)
Te Kowhai – wet	90	82–99 (2)
Te Kowhai – dry	87	82–92 (2)
Taupo – wet	89	83–94 (2)
Taupo – dry	93	90–95 (2)

Table 3 Depth distribution of urine Br (means for normalised data). Where Br > 20 cm, between site variance (s.e.) = 3.2 (1.4), within-site variance = 1.1 (0.3). Where Br > 30 cm, between site variance (s.e.) = 2.5 (1.0), within-site variance = 2.5 (0.1).

Depth	Waikiwi	Pukemutu	Mokotua	Fleming drained	Fleming undrained	Temuka	Lismore	Silverdale wet	Silverdale dry	Whangaripo wet	Whangaripo dry	Te Kowhai wet	Te Kowhai dry	Taupo wet	Taupo dry
0–10	63	81	73	87	79	52	68	80	68	59	23	46	31	95	95
10–20	22	10	14	7	10	23	23	16	18	14	10	14	24	5	5
20–30	11	7	5	3	4	25	9	3	7	11	13	13	11		
30–40	3	2	3	2	7			1	4	7	22	9	14		
40–50	2		3	1					2	5	19	8	9		
50–60	1		4						1	3	9	6	5		
60–70			1						1	2	4	3	5		
70–80												1	2		
Br > 20 cm	15	9	14	6	11	25	9	4	14	28	68	40	46	0	0
Br > 30 cm	4	2	9	2	4	0	0	1	7	17	55	27	35	0	0

analysis of the data. The very high flows of solution to beyond 20 cm noted for the Whangaripo and Te Kowhai soils examined in this study are similar in magnitude to those reported by Williams et al. (1990a), who measured losses of 48% of the urine applied to a Taupo soil in winter, and up to 74% of urine applied to an undisturbed lysimeter containing Manawatu soil. Losses, however, were concluded to be more dependent on soil surface physical conditions than soil chemical and mineral characteristics or soil moisture contents.

Relationships between Br distribution and soil physical properties

K_S

It was anticipated that good relationships would exist between Br movement and measurements of soil hydraulic conductivity and air permeability. However, regression analysis of Br movement below 20 cm and K_S indicated that these relationships were generally poor, both when data were analysed collectively (all urine patches and soil types), or when analysed separately within individual soil types. Correlations were initially made using mean (geometric), maximum, or minimum values of K_S measured within each set of three cores taken from each soil layer. Mean values of K_S were, in almost all cases, found to have higher correlation coefficients than maximum values, which in turn were higher than minimum values. For this reason, the results presented here are limited to correlations using mean values for K_S for each soil layer.

Correlation coefficients between Br movement below 10, 20, or 30 cm and measurements of K_S at various soil depths are shown in Table 4. These correlations were carried out using either the combined data set from all of the site replicates examined (All data) or mean values calculated for each of the sites examined (Site means). Although many of the correlations in the combined dataset proved to be statistically significant, they accounted for less than 50% of the variation observed between Br movement and K_S. The relationships between Br movement and K_S were even poorer when correlations were calculated within each soil type (data not shown). Surprisingly, no significant relationships were found between Br movement below 20 cm (or 30 cm) and K_S measured in the 0–5 cm soil layer; indeed correlations between Br movement and K_S for this soil layer were poorest overall.

In contrast to the poor relationships when all data were combined, improved relationships between K_S and Br movement resulted from correlations carried out on mean values calculated for each of the 11 sites (10 soils) examined in this study. Once again, poor correlations were noted for K_S measured at 0–5 cm, but improved and significant correlations were noted between Br movement and K_S measured in deeper soil layers, particularly at 20–30 and 30–40 cm. The highest correlations with Br movement were observed for the averaged 0–30 cm K_S values (Fig. 1), although it is apparent that these higher correlations arise mainly because the data are clustered in two distinct

Table 4 Correlation coefficients ($r \times 100$) for relationships between Br movement and K_S, geometric mean of the 3 cores taken per depth. Correlations performed using either the combined data set from all of the site replicates examined (n = 43) or mean values calculated for each of the sites examined (Site means, n = 11). *, $P < 0.05$; **, $P < 0.01$.

Depth (cm)	Br > 10 cm All data	Br > 10 cm Site means	Br > 20 cm All data	Br > 20 cm Site means	Br > 30 cm All data	Br > 30 cm Site means
0–5	3	<1	21	38	26*	50
5–10	42**	61*	51**	74**	51**	68*
10–20	48**	69*	51**	72**	53**	70*
20–30	43**	88**	53**	91**	57**	86**
30–40	25	95**	23	91**	19	88*
0–10[a]	24	31	36*	55*	39**	60*
0–20[b]	50**	72**	60**	88**	63**	89**
0–30[c]	57**	88**	67**	96**	70**	92**

[a]Mean of 0–5 and 5–10 cm layers.
[b]Mean of 0–10 and 10–20 cm layers.
[c]Mean of 0–10, 10-20, and 20–30 cm layers.

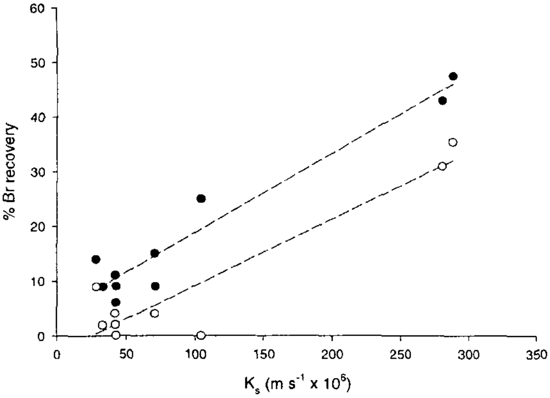

Fig. 1 Relationship between Br recovery below 20 cm (●) or 30 cm (○) and mean soil K_S, average of 0–30 cm layer.

groups. The first cluster represents the Te Kowhai and Whangaripo soils, which exhibit a high degree of preferential flow and high K_S values. The second cluster of data represents the remaining soils, which, on average, had a preferential flow to below 20 cm of only 11% of the urine applied and K_S values generally less than 10^{-4} m s^{-1}.

The improvement in statistical correlations between Br movement and mean K_S values using deeper soil layers suggests that either variability in K_S was lower at deeper soil layers, or the limiting factor for Br movement to depth was the hydraulic conductivity of subsoil layers which could potentially restrict the movement of urine from above. Values for K_S measurements in the 0–5 cm layer varied considerably, and, in hindsight, 3-fold replication of cores for estimating K_S for this soil layer proved to be inadequate. Soil physical properties within this surface layer are known to vary greatly both spatially and temporally as a result of such processes as compaction by animal treading and burrowing by earthworms. This variability is reflected in the Coefficients of Variation calculated using log-transformed K_S values: the averages for all soils examined were 39, 37, 26, 33, and 26% for the 0–5, 5–10, 10–20, 20–30, and 30–40 cm layers, respectively. The improved correlations between mean values for soils reflect the better estimates of K_S obtained for a particular soil when an increased number of replicate cores were used for estimating mean K_S values. The use of larger soil cores for determining K_S would also have improved estimates of K_S.

Visual observations of pyranine dye within lower soil depths indicated that much of the urine was moving to depth via earthworm channels and, in some situations, via cracks in the soil. In the case of the Mokotua soil, dye was also noted in an old large root channel. Urine movement through these macropores led to an obvious fingering pattern in almost all of the soils examined, with the exception of Taupo. Measurement of K_S under such a random fingering pattern, using the small cores used in this study, is difficult. In retrospect, in-situ K_S measurement would have been preferable to the core approach used here, ideally at the same location at which urine is later applied. Increased replication using the small core approach may have improved the relationship between Br movement and Ks, although the replication required may have proven too great to be a practical option for estimating field K_S.

Table 5 Correlation coefficients ($r \times 100$) for relationships between Br movement and K_a, geometric mean of the 3 cores taken per depth. Data for the six South Island soils only. Correlations performed using either the combined data set from all of the site replicates examined ($n = 23$) or mean values calculated for each of the sites examined (Site means, $n = 7$). *, $P < 0.05$; **, $P < 0.01$.

Depth (cm)	Br > 10 cm All data	Br > 10 cm Site means	Br > 20 cm All data	Br > 20 cm Site means	Br > 30 cm All data	Br > 30 cm Site means
0–5	43*	49	29	40	29	0
5–10	54**	69	37*	51	0	54
10–20	61**	75*	56**	83*	0	32
20–30	66**	98**	58**	90**	0	0
0–10[a]	55**	72*	38*	58	0	0
0–20[b]	63**	82*	50**	79*	0	0
0–30[c]	68**	89**	56**	84*	0	0

[a]Mean of 0–5 and 5–10 cm layers.
[b]Mean of 0–10 and 10–20 cm layers.
[c]Mean of 0–10, 10–20, and 20–30 cm layers.

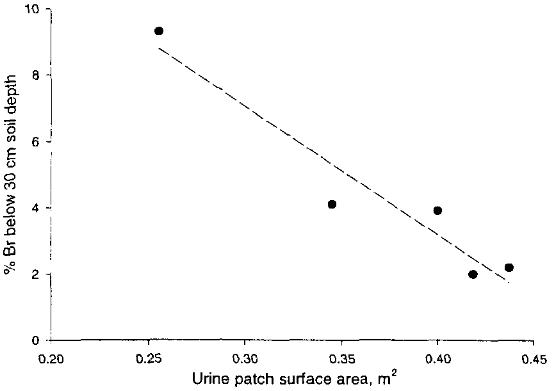

Fig. 2 Relationship between urine patch surface area and Br movement below 30 cm. Plotted points are mean values for the five Southland sites (four soil types) examined. Bromide below 30 cm = 18.6–38.6*surface area.

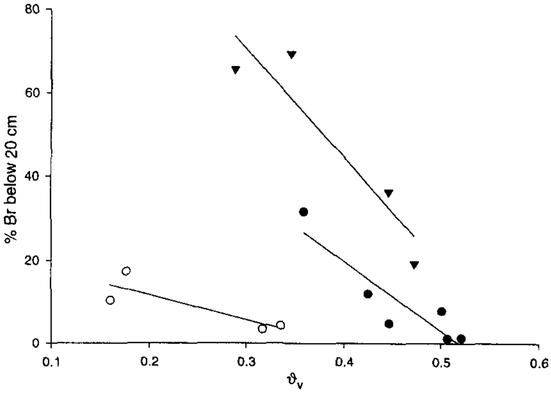

Fig. 3 Relationship between Br movement below 20-cm soil depth and soil volumetric water content (θ_v) (0–20 cm) for the Pukemutu (•), Silverdale (○), and Whangaripo (▼) soils.

K_a, urine patch surface area, and θ_v

As for K_S, K_a measurements made on the cores taken from the six South Island soils were highly variable, particularly at 0–5 and 5–10 cm. Geometric mean values, as opposed to maximum or minimum values, were again found to be better correlated with Br movement, although correlations using the combined South Island dataset were again quite poor (Table 5) and generally lower than those found for K_S when Br movement below 20 or 30 cm was examined.

There was a very poor correlation between urine patch surface area and Br movement with depth ($r < 0.2$) for the six South Island soils, using data from individual urine patches. Poor correlations between Br movement below 10 and 20 cm soil depth and urine patch surface area also occurred from mean values for each soil examined. There was, however, a significant ($P < 0.05$) relationship between mean urine patch surface area and mean Br movement below 30 cm for the six South Island soils (Fig. 2).

Clear relationships between soil volumetric water content and Br movement below 20 cm were evident for the dry and wet samplings of the Silverdale, Pukemutu, and Whangaripo soils only (Fig. 3). At lower soil moisture contents, greater amounts of Br moved below 20 cm in these three soils. There did not appear to be any significant relationship between soil moisture content and Br movement in the Te Kowhai soil, however, which was also sampled under both wet and dry conditions. When correlations were made using data from all urine patches, there was no clear relationship between soil moisture content and Br movement, indicating that soil moisture effects were soil- or site-specific. Similarly, there was no clear relationship between Br movement below 20 or 30 cm and soil aggregate size classes (data not shown).

In summary, the use of other soil physical measurements such as moisture content and aggregate size classes, air permeability, and urine patch surface area did not account for any more of the observed variability than that obtained using K_S. The ability to predict preferential flow of urine based on soil moisture content or K_S obtained using the small cores employed in this study was thus generally poor.

Preferential flow versus leaching losses of K

For the two soils where concurrent measurements of K intake and losses in drainage water were being measured (Silverdale and drained Fleming soils), the amounts of K lost via preferential flow of urine to below 20 cm exceeded the K losses measured in drainage waters (Table 6). Reported K losses in drainage water are the mean annual losses measured over 3 years for the Silverdale soil and over 2 years for the Fleming soil. Relatively low amounts of K leaching from pastures have also been noted in studies by Smith et al. (1984; 4.4 kg $ha^{-1}yr^{-1}$) and Steele et al. (1984; 14 kg $ha^{-1}yr^{-1}$) and from large undisturbed lysimeters treated with cow urine (Early et al. 1998). Williams et al. (1990c, 1990d) also noted that preferential flow losses of K

Table 6 Losses of K via preferential flow of urine and drainage for Southland and Waikato pastures grazed by cattle. For the Waikato study, 65 kg K ha^{-1} was estimated to have been transferred from pasture to the milking shed and lanes. K deposition in urine assumes 81% of K ingested is excreted in urine (Hutton et al. 1967). Preferential flow uses values from Table 3, with mean of "wet" and "dry" samplings for the Silverdale soil.

Location Soil type	Southland Fleming	Waikato Silverdale
K intake, kg ha^{-1}	400	469
K deposition in urine, kg ha^{-1}	324	327
Preferential Flow below 20 cm, kg ha^{-1}	19	29
Preferential Flow below 30 cm, kg ha^{-1}	6	13
K lost in drainage, kg ha^{-1}	10	8

exceeded those estimated to be lost via drainage. The balance may be accounted for by the retention of K on soil colloids, or perhaps by the uptake of K by roots present below 20 cm. Measurements of exchangeable K to 50 cm soil depth in the Silverdale soil did show a considerable amount of K below 20 cm, suggesting that K retention in the subsoil may be significant (Table 7). In contrast, measurements of exchangeable K levels below 20 cm for the drained Fleming soil were quite low and did not indicate a significant accumulation of K in the subsoil. The Fleming soil site had been grazed by cattle for only 3 years prior to the measurements of exchangeable K being made, and hence may not provide a true picture of soil K distribution under established dairy pastures, whereas the Silverdale soil site had been under dairying for more than 30 years.

Under field conditions, most of the nutrient uptake actually occurs from the 0–10 cm soil layer, reflecting the high levels of plant-available nutrients which are to be found in this layer in pastoral soils. However, provided there are no physical impediments, some roots of most pasture species penetrate to considerable depth. Evans (1978), in a Karapoti brown sandy loam with swards of white clover, perennial ryegrass, a white clover-ryegrass mixture, or cocksfoot, found 70–80% of roots in the top 20 cm, with the remaining 20–30% showing an exponential decline in root length density down to 1.4 m. Harrison et al. (1994), in a mixed pasture on a Templeton silt loam, found a near-linear decline in root length density to 50 cm with some roots at 80 cm. Similar results have been found for a wide range of pasture species in Australia (Ozanne et al. 1965). Plant uptake of radioactive K (Ozanne et al. 1965) placed at different soil depths has been shown to be related to the quantity of root present at that depth. Studies with P (Jackman & Mouat 1972; Gillingham et al. 1980; Syers et al. 1984) and S (Gregg et al. 1977; Harrison 1993) tracers have also shown plant availability of these nutrients placed at various depths in the profile, although relative activity may also reflect nutrient distribution in the profile.

Table 7 Soil exchangeable K levels ($cmol_c\ kg^{-1}$) for the Southland and Waikato pastures grazed by cattle.

Depth cm	Southland Fleming soil	Waikato Silverdale soil
0–10	0.33	0.49
10–20	0.16	0.26
20–30	0.08	0.22
30–40	0.05	0.20
40–50	0.03	0.19

A mean distribution of urine K in the soil profile was calculated from eight sites (seven soils) examined in this study, excluding the Taupo soil with no evidence of macropore flow and the two soils (Whangaripo and Te Kowhai soils) with high macropore flow, and an exponential decay curve fitted

$$y = 12.2\exp(-0.122x) \qquad r^2 = 0.99 \qquad (1)$$

where y is urine K recovery (%) per cm soil depth and x is soil depth (cm). Similarly, a relative root distribution curve was fitted to the data of Harrison (1993) and Evans (1978) (Fig. 4). A comparison of this root distribution curve with the mean urine distribution curve calculated for the eight sites with moderate macropore flow suggests that there is likely to be a greater proportion of root mass in the 10–50 cm layer than urine-derived nutrients. Although using a generalised root distribution curve

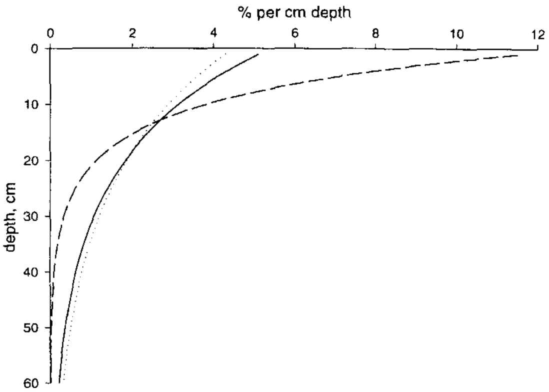

Fig. 4 Relative distributions of urine K and plant roots in soil. —— Relative root distribution, from Harrison (1993) and Evans (1978); • • • • K distribution, Whangaripo & Te Kowhai soils; - - - - K distribution, other soils.

is speculative, the comparison does suggest that, in these soils, macropore flow of urine is unlikely to result in an immediate loss of nutrients from the plant available pool.

A mean distribution of urine K was also calculated for the two soils with high macropore flow (Te Kowhai and Whangaripo, Fig. 4). Surprisingly, the relative urine K distribution was almost identical to the relative root distribution obtained using the data of Harrison (1993) and Evans (1978). For depths greater than 20 cm there was a slight surplus of urine nutrients relative to root abundance, with the proportion of nutrients below 20 cm exceeding the proportion of roots below 20 cm by about 6%. Again, although speculative, the comparison does suggest that a loss of urine-derived nutrients is more likely than in the eight other soil types that displayed only a moderate degree of macropore flow. However, it is noteworthy that maintenance K fertiliser requirements are similar for dairy pastures on the Silverdale and Te Kowhai soils (A. Roberts pers comm.), indicating that K loss processes are of equal magnitude for these two soil types despite the higher degree of preferential flow in the latter.

These results suggest that macropore flow of cow urine to depths below 20 cm cannot be regarded as an immediate loss of nutrients from the soil-plant-animal system. For nine of the soils examined in this study, although macropore flow of urine to below 20 cm soil depth was evident, it is likely that root uptake would recover all or most of the urine nutrients below 20 cm. Even for the Whangaripo and Te Kowhai soils, which exhibited a high degree of preferential flow, relative root abundance at lower soil depths is still likely to be sufficient to recover much of the urine nutrients.

CONCLUSIONS

The proportion of simulated cow urine moving via preferential flow to below 20 cm soil depth ranged from 0 to 68% of that applied, and averaged 17% for the 10 soil types examined in this study. Two soils, Whangaripo and Te Kowhai, exhibited a high degree of preferential flow; their exclusion from the dataset reduced this average flow to 11%. The preferential flow of urine to below 20 or 30 cm soil depth was best predicted by K_s measurements, as determined for the 0–30 cm soil layer. K_s and K_a measurements made for any one 5-cm soil depth increment, particularly the 0–5 cm soil depth, were poor predictors of preferential urine flow. Although the preferential flow of urine to soil depths below 20 cm was significant, it does not represent a major loss pathway of urine K based upon relative root distributions calculated for a typical New Zealand pasture.

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